Fiscal Year:	FY 2017	**Task Last Updated:**	FY 11/16/2017
PI Name:	Eisch, Amelia Ph.D.		
Project Title:	HZE Particle Exposure-Induced Improvement of Pattern Separation in Mature Mice: Alterations in Mission-Relevant Behaviors and Neural Circuitry		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104	**Congressional District:**	3
Comments:	NOTE: Previously at University of Texas Southwestern Medical Center at Dallas, 2000-2016		
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	01/30/2015	**End Date:**	07/17/2017
No. of Post Docs:	4	**No. of PhD Degrees:**	1
No. of PhD Candidates:	9	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/17/2017 per S. Monk/HRP, due to PI move to Childrens' Hospital of Philadelphia; original end date was 1/29/2019 (Ed., 8/15/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Yun, Sanghee Ph.D. (Children’s Hospital of Philadelphia)		
Grant/Contract No.:	NNX15AE09G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	An unavoidable consequence of deep space missions is exposure to galactic cosmic radiation (GCR), which includes high (H) atomic number (Z) and energy (E) particles particles like Fe, Si, and O. Estimating radiation risks to the central nervous system (CNS) by HZE particles encountered during space missions is a high research priority. Past research has shown that rodents exposed to HZE particles have cognitive and performance deficits in numerous behavioral tasks, including those that rely on the hippocampus, a brain region involved in learning and memory. Notably, we have found that mature mice (of equivalent age to astronauts) exposed to either Si or Fe HZE particles actually show improved performance on a very difficult hippocampal task to assess the ability to discrimination two contexts that differ in discrete ways. Here we propose three aims to understand this improved context "pattern separation" after HZE particle exposure. In Aim 1, we hypothesize that HZE particle exposure-induced improved pattern separation is linked to improved performance on related learning and memory tasks, as well as executive function tasks, in the short-term, but to decreased performance in the long-term. In Aim 2, we hypothesize that HZE particle exposure-induced improved pattern separation is linked in the short-term to diminished stress-induced emergence of anxiety and depression-like behaviors, but to greater emergence in the long-term. In Aim 3, we hypothesize that HZE particle exposure-induced improved pattern separation is associated with disrupted hippocampal-cortical neural networks. All aims will rely on both classic and cutting-edge techniques. In sum, these aims will address whether the HZE particle exposure-induced improvement in pattern separation is beneficial or detrimental to mission success (Aims 1, 2), will indicate the integrity of neural circuitry contributing to mission-relevant behaviors (Aim 3), and will define both the short- and long-term health of neural networks needed to complete deep space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An unavoidable aspect of manned space flights is exposure to galactic cosmic radiation (GCR), which is made up primarily of protons (87%), followed by helium (11%) and then high atomic number (Z) and high-energy (HZE) particles (2%) like iron, silicon, and oxygen. The energy of HZE particles can be very high (>1000 MeV/u), sufficient in many cases to penetrate the spacecraft hull and interior materials, and they have a complex track structure and high linear energy transfer (LET). Thus, while the greatest physical radiation dose of GCR comes from high-energy protons, the greatest biological radiation dose of GCR comes from HZE particles. With long-duration and exploratory space missions in the near future, we need to understand how GCR influences human health and behavior. Estimation of radiation risks to the central nervous system (CNS) is a high research priority according to both a National Academy of Science report and NASA's Radiation Health Bioastronautics Roadmap. In rodents, HZE particles induce cognitive domain deficits, including decreased hippocampal learning and memory and cortically based executive function. However, it remains unclear whether the age at irradiation (IRR) influences the outcome of behavioral tests. It is also unclear whether these cognitive decrements extend to other cognitive tests, or to other behavioral domains, such as mood and stress response. Recently, we found that mice exposed at maturity ("astronaut-aged") to 28Si or 56Fe HZE particle IRR perform better than control mice on a hippocampal-based pattern separation task (context discrimination fear conditioning, CDFC). We want to understand if this behavioral improvement in pattern separation is reflective of other changes in behavior, and whether these changes will be beneficial or detrimental to mission success. We propose a behavioral domain- and brain network-based analysis to understand the HZE particle-induced behavioral improvement shown in our pilot data. These data will have relevance for understanding the risks facing crew members in deep space missions, particularly in regards to the age of crew member at the time of the mission.
Task Progress:	Using mice that irradiated at astronaut-age and multiple time points post-IRR, we have found that mice of astronaut age show improved pattern separation in both appetitive and aversive tasks. We have also established an ethologically-relevant stressor, social defeat stress, in our new institution (PennMed/CHOP) and we hypothesize that improved pattern separation is linked to resilience to social stressors. Finally we used RNA-seq to follow upon work examining the translationome of genetically-defined granule cells to identify key pathways altered after radiation. Publications supported by this grant: 1. Latchney SE*, Jiang Y*, Petrik D, Eisch AJ^, Hsieh J^. Inducible knockout of Mef2A/C/D from nestin-expressing stem/progentior cells and their progeny unexpectedly uncouples neurogenesis and dendritogenesis in vivo. FASEB J. 2015 Dec;29(12):5059-71. Epub 2015 Aug 18. http://dx.doi.org/ ; PubMed PMID: 26286136 ; PubMed Central PMCID: PMC4653059 ; *equal contribution, ^co-corresponding authors 2. Petrik D, Latchney SE*, Masiulis I*, Yun S*, Zhang Z*, Wu JI^, Eisch AJ^. Chromatin remodeling factor Brg1 supports the early maintenance and late responsiveness of nestin-lineage adult neural stem and progenitor cells. Stem Cells. 2015 Dec;33(12):3655-65. Epub 2015 Oct 9. http://dx.doi.org/ ; PubMed PMID: 26418130 ; PubMed Central PMCID: PMC4713255 . *equal contribution, listed alphabetically, ^co-corresponding authors 3. Whoolery CW, Walker AK, Lucero MJ, Richardson DR, Reynolds RP, Beddow DH, Clark KL, Shih H-Y, LeBlanc JA, Cole MG, Amaral WZ, Mukherjee S, Zhang S, Ahn F, Bulin SE, DeCarolis NA, Rivera PD, Chen BPC, Yun S, Eisch AJ. Whole Body Exposure to 28Si Radiation Dose-Dependently Disrupts Dentate Gyrus Neurogenesis and Proliferation in the Short-Term and New Neuron Survival and Contextual Fear Conditioning in the Long-Term. Radiat Res. 2017 Nov;188(5):532-551. Epub 2017 Sep 25. https:// ; PubMed PMID: 28945526 4. Ochoa Escamilla C*, Filonova I*, Walker AK, Xuan Z, Holehonnur R, Espinosa F, Liu S, Thyme SB, Lopez-Garcia IA, Mendoza DB, Usui N, Ellegood J, Eisch AJ, Konopka G, Lerch JP, Schier AF, Speed HE, Powell CM. Kctd13 deletion reduces synaptic transmission via increased RhoA. Nature. 2017 Nov 9;551(7679):227-31. https:// ; PubMed PMID: 29088697 , *equal contribution 5. Norbury JW, Schimmerling W, Slaba TC, Azzam E, Badavi FF, Baiocco G, Benton E, Bindi V, Blakely EA, Blattnig SR, Boothman DA, Borak1 TB, Britten RA, Curtis S, Dingfelder M, Durante M, Dynan W, Eisch AJ, Elgart SR, Goodhead DT,

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7. Yun S, Reynolds RP, Masiulis I, Eisch AJ. Re-evaluating the link between neuropsychiatric disorders and dysregulated adult neurogenesis. Nature Medicine, 22(11):1239-1247 (2016). PubMed PMID: 27783068 , PMCID in progress (NIHMSID 888041).

NOTE (Ed., 11/20/17): Project continues as "HZE Particle Exposure-Induced Improvement of Pattern Separation in Mature Mice: Alterations in Mission-Relevant Behaviors and Neural Circuitry--80NSSC17K0060" with the same Principal Investigator, Dr. Amelia Eisch, due to PI move to Childrens' Hospital of Philadelphia ; see that project for subequent reporting.

Bibliography Type:	Description: (Last Updated: 11/17/2025)
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Articles in Peer-reviewed Journals	Norbury JW, Schimmerling W, Slaba TC, Azzam EI, Badavi FF, Baiocco G, Benton E, Bindi V, Blakely EA, Blattnig SR, Boothman DA, Borak TB, Britten RA, Curtis S, Dingfelder M, Durante M, Dynan WS, Eisch AJ, Robin Elgart S, Goodhead DT, Guida PM, Heilbronn LH, Hellweg CE, Huff JL, Kronenberg A, La Tessa C, Lowenstein DI, Miller J, Morita T, Narici L, Nelson GA, Norman RB, Ottolenghi A, Patel ZS, Reitz G, Rusek A, Schreurs AS, Scott-Carnell LA, Semones E, Shay JW, Shurshakov VA, Sihver L, Simonsen LC, Story MD, Turker MS, Uchihori Y, Williams J, Zeitlin CJ. "Galactic cosmic ray simulation at the NASA Space Radiation Laboratory." Life Sciences in Space Research. 2016 Feb;8:38-51. https://doi.org/10.1016/j.lssr.2016.02.001 ; PubMed PMID: 26948012 , Feb-2016
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